

Novel Autophagy Modulators in Neurodegenerative Disease: Beyond Amyloid Pathways"

Abstract: Alzheimer's Disease (AD), a debilitating neurodegenerative disorder, has primarily been targeted through amyloid-beta and tau pathology. However, recent evidence points towards a significant role of lysosomal and autophagic dysfunction in neuronal degradation. This paper explores the intricate mechanisms of autophagy dysregulation in AD progression, focusing on pathways involving specific genes such as *ATG7* and *ULK1*, and their downstream effects on cellular waste clearance. We present findings from in-vitro studies demonstrating that certain small molecules, not traditionally associated with neuroprotection (e.g., **Rapamycin**), can modulate autophagy flux and reduce aggregate accumulation in neuronal cell lines. Furthermore, preliminary observational data from patients with other conditions (e.g., organ transplant recipients) on chronic low-dose Rapamycin have shown unexpected, albeit anecdotal, improvements in cognitive metrics. This suggests a potential avenue for drug repurposing, targeting autophagic pathways as a novel therapeutic strategy for neurodegenerative diseases, independent of direct amyloid or tau targeting.

Introduction: Alzheimer's disease is a complex multifactorial disorder. While amyloid plaques and neurofibrillary tangles are hallmarks, therapeutic successes have been limited. Emerging research highlights impaired cellular clearance mechanisms, particularly autophagy, as critical contributors to neurodegeneration. Autophagy is a catabolic process involving the degradation and recycling of cellular components. Dysfunction in this pathway leads to the accumulation of misfolded proteins and damaged organelles.

Autophagy Genes and AD: Genetic studies have implicated variations in autophagy-related genes (e.g., *ATG5*, *ATG7*, *ULK1*, *BECN1*) in AD risk and progression. These genes play crucial roles in autophagosome formation and maturation. Specifically, decreased expression of *ATG7* has been linked to impaired amyloid precursor protein (APP) processing.

Rapamycin as an Autophagy Inducer: Rapamycin, a known mTOR inhibitor and immunosuppressant, has been extensively studied for its autophagy-inducing properties. While its primary clinical use is in preventing transplant rejection, its ability to enhance lysosomal function and clear aggregated proteins in neuronal models suggests a broader therapeutic potential. Our in-vitro experiments show that Rapamycin treatment significantly reduces levels of aggregated tau and amyloid-beta oligomers in AD patient-derived iPSC neurons.

Clinical Observations and Repurposing Hypothesis: Anecdotal reports and small observational studies of transplant patients on chronic Rapamycin therapy have occasionally noted preserved or improved cognitive function compared to age-matched

controls not on the drug. This observation, combined with the strong mechanistic rationale, generates the hypothesis that **Rapamycin, or other specific mTOR inhibitors**, could be repurposed for the treatment of Alzheimer's Disease, targeting the underlying autophagy deficits. Further clinical trials are warranted.

Conclusion: Modulating autophagy presents a promising, yet underexplored, therapeutic strategy for Alzheimer's Disease. Repurposing existing drugs like Rapamycin, which already have established safety profiles, could accelerate the development of much-needed treatments.